DO NOW — RECALL

GPUs, multicore and parallel systems

Name: _____ Class: _____ Date: _____

1. Why doesn't a smaller assembly instruction count guarantee a faster program?

2. What is a core?

3. Two independent CPU-bound tasks each take 4 minutes when run on its own core. They start together on two separate identical cores. Calculate the ideal elapsed completion time, ignoring overhead.

4. Given int[] a={1,2,3,4}, perform a[i]=a[i]*2 for even indexes 0 and 2 only. Give final array and its sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Different instructions have different costs, and decoding, caches and pipelining affect performance.

2. Model answer

An execution unit capable of carrying out its own instruction stream.

3. Model answer

4 minutes. Both cores execute at the same time and each finishes its task after 4 minutes.

4. Model answer

Final [2,2,6,4], sum=14. Indexes 0 and 2 were changed, not the even-valued elements.

Vocabulary — know these terms

GPU — Processor designed for highly parallel workloads.

multicore — CPU containing multiple execution cores.

data parallelism — Apply the same operation to many independent data elements.

bandwidth — Volume of data transferred in a given time.

synchronisation — Coordinating shared work and results.

NEW LEARNING

Read, annotate and explain

GPUs and data parallelism

A GPU contains many execution resources suited to repeating similar calculations across graphics pixels or machine-learning tensors.

Multicore CPUs

Multiple CPU cores can run independent threads/processes, enabling greater throughput when tasks can be parallelised.

Evaluate overhead

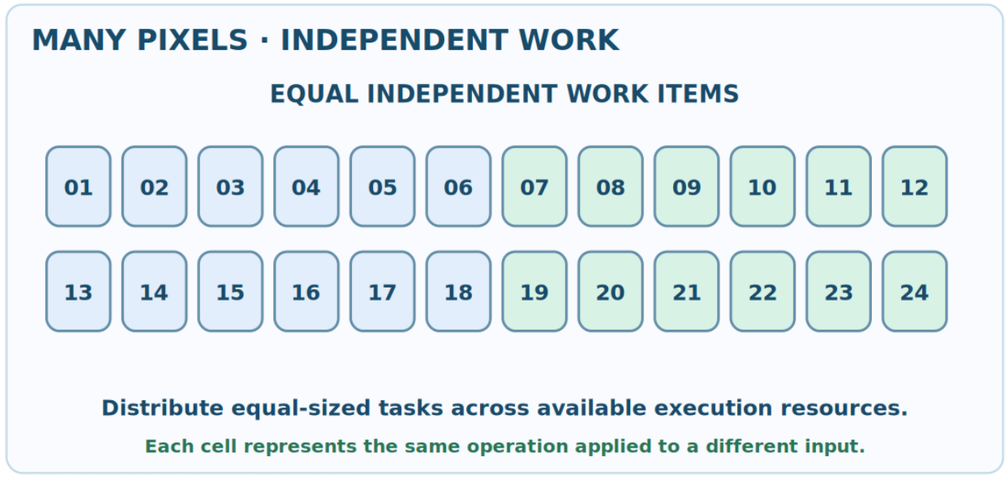
Data transfers, branching, synchronisation and sequential tasks can reduce the ideal speedup. A GPU is not automatically best for every problem.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

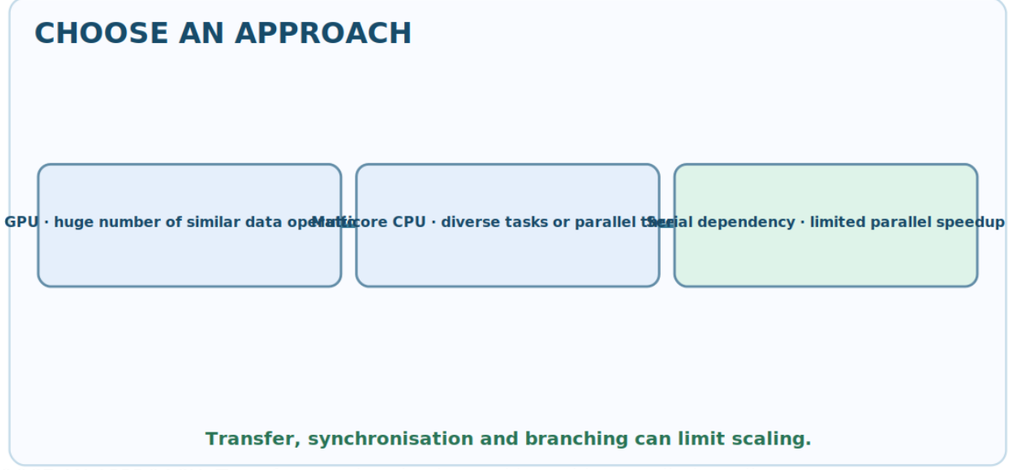
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



MANY PIXELS · INDEPENDENT WORK: Each cell represents the same operation applied to a different input.

New Learning



CHOOSE AN APPROACH: Transfer, synchronisation and branching can limit scaling.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Each core receives four blocks; coordination overhead and data movement may affect real runtime.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

GPUs always execute any sequential algorithm faster than a CPU.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

Compare 12 independent image tiles split across four cores under an ideal equal-work model.

Predict first

Divide 12 independent equal-sized tiles among four available cores.

Explain and check

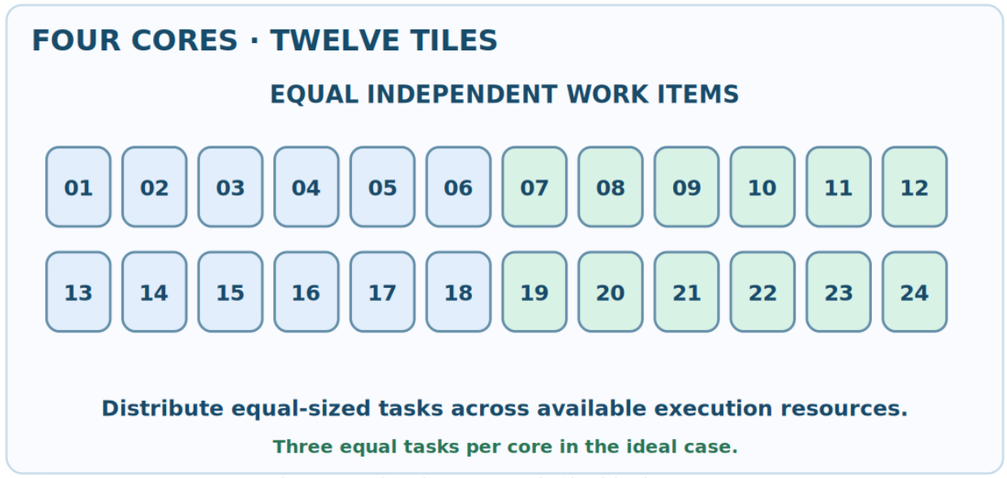
Each core processes three tiles, so ideal completion is around one quarter of serial time only if work and overhead allow.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare 12 independent image tiles split across four cores under an ideal equal-work model.

Divide 12 independent equal-sized tiles among four available cores.



FOUR CORES · TWELVE TILES: Three equal tasks per core in the ideal case.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A video-editing workflow must first decode frames sequentially, then apply the same filter independently to each frame, before assembling output. Explain which parts suit GPU execution and why.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give two uses of GPUs, including one not based on graphics, and explain a common characteristic of both.

Question 2

A task has 24 independent identical chunks and six CPU cores. Calculate ideal work per core and describe two constraints on real speedup.

WE DO — GUIDED PRACTICE

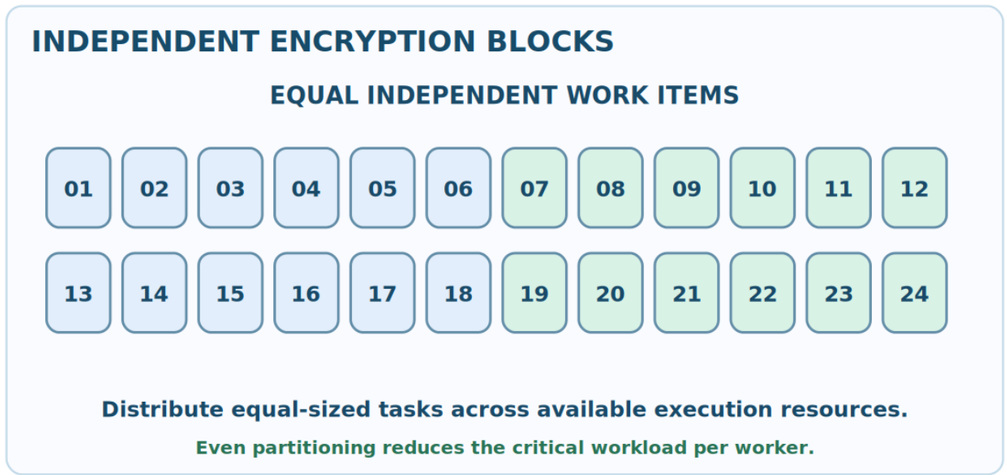
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

An encryption batch has 20 independent blocks and five worker cores. How many blocks per core ideally?

The blocks can be divided evenly into five independent groups.



INDEPENDENT ENCRYPTION BLOCKS: Even partitioning reduces the critical workload per worker.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The blocks can be divided evenly into five independent groups.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.